

THE EDGE-BOREL GROWTH CONSTANT IS A
NON-CLASSICAL EXPONENTIAL PERIOD: AN SL_2
MONODROMY ON THE PAINLEVÉ III(D_8) SURFACE
AND A CONDITIONAL TRANSCENDENCE THEOREM

PAPANOKECHI

Date: June 14, 2026.

2020 Mathematics Subject Classification. 34M55, 34M40, 34M03, 11J81, 11G09, 33E17.

Key words and phrases. isomonodromy, Painlevé III, Sakai surface, exponential period, connection coefficient, Stokes constant, differential Galois group, resurgence, conditional transcendence, polynomial continued fraction.

Independent Researcher (ORCID 0009-0000-6192-8273). This is a preprint; nothing herein is a claim of priority. Every load-bearing statement is graded PROVEN/STRUCTURAL/VERIFIED/CONJECTURED (§1.2); PROVEN is reserved for machine-checked Lean 4 cores, of which this paper introduces none (the program substrate is the EBR-III cores, §11).

ABSTRACT. For the positive-coefficient polynomial continued fraction with $b(n) = 3n^2 + n + 1$, the Edge–Borel growth constant is governed by a single transcendental number $\kappa = 1.5394948485766410\dots$, the Stokes/connection constant of a rank-2 ordinary differential operator H_2 that arises by a Borel–2 reduction of the generating function. We identify κ inside the isomonodromy and period frameworks and assemble, with explicitly graded hypotheses, a conditional transcendence theorem for it. First, H_2 has singular set $\{0, \infty\}$ with both points irregular of ramified slope $\frac{1}{2}$; its index of rigidity is $\text{rig}(H_2) = 0$ (non-rigid, moduli dimension 2), and the local ramification data $(2, 2)$ select, under a surface-type selector we state precisely, the Sakai surface $D_8^{(1)}$ —equivalently $P_{\text{III}}(D_8)$. Second, the differential Galois group is $G_{\text{Gal}}(H_2) = \text{SL}_2$ (an exact Kovacic computation), so H_2 is irreducible and has no Liouvillian solutions. Third, on the convergent Borel–2 transform Φ the constant κ is realised *constructively* as the connection coefficient A_Φ of an order-4 operator with an irregular point of slope $\frac{1}{4}$, computed by a monodromy spectral projector to 129 digits and agreeing with two further independent channels; this places κ in the ring of exponential (rapid-decay) periods of the connection. We determine the monodromy point $(\text{tr } M_0, \kappa)$ with $\text{tr } M_0 = -51.0655631399546\dots$ (hyperbolic, hence irreducible), survey the solved D_8 connection problems and find that none computes κ (a tau-side/Lax-side distinction we keep explicit), and record two positive-control-validated integer-relation nulls. The point lies off the algebraic $P_{\text{III}}(D_8)$ locus. Under the Fresan–Jossen period conjecture for exponential motives, a differential-to-motivic dimension comparison, and a verified non-degeneracy datum, we conclude $\kappa \notin \mathbb{Q}$, hence the EBR connection coefficient $C = \kappa\sqrt{\pi}/\Gamma(4/3)$ is transcendental. We are explicit, throughout and in both directions, that this is a conditional statement: neither the large Galois group nor the non-rigidity nor any null proves transcendence, and a closed form—had one appeared—would have argued the opposite.

1. INTRODUCTION

1.1. The program and the object. This paper continues the Edge–Borel Radius (EBR) program. EBR-I established that for a positive-coefficient polynomial continued fraction $Q_n = b(n)Q_{n-1} + Q_{n-2}$ with $b(n) = \beta_d n^d + \dots$, the Borel object $G(s) = \sum_{n \geq 0} Q_n s^n / (dn)!$ is holonomic with a dominant singularity at $s = R = d^d / \beta_d$ and local form $G(s) \sim A(1 - s/R)^{-\gamma}$ [1]. EBR-II identified the amplitude A as a *connection coefficient* of the order-2d operator annihilating G and conjectured it to be generically transcendental [2]. EBR-III fixed the smallest non-rigid case $d = 2$, $b(n) = 3n^2 + n + 1$, and proved (structurally, with audited Lean cores) that the order-4 operator there is not a G -operator and has differential Galois group SL_4 ; it left the transcendence of the connection coefficient explicitly open as the burden of a separate period analysis [3].

This paper carries out that period analysis for $d = 2$. Its object is the single constant

$$\kappa = 1.53949484857664103484378190338406903821939\dots, \quad (1)$$

the Stokes/connection constant controlling the Edge–Borel growth, related to the EBR-I/II amplitude by the exact elementary identities

$$\kappa = \Gamma(\tfrac{4}{3}) A_0, \quad A_0 = \frac{C_{\text{EBR}}}{\sqrt{\pi}}, \quad C = C_{\text{EBR}} = \frac{\kappa\sqrt{\pi}}{\Gamma(4/3)}, \quad (2)$$

where $A_0 = \lim_n Q_n / ((n!)^2 3^n n^{1/3})$ and $C_{\text{EBR}} = 3.0557068078904813\dots$ is the EBR-III connection coefficient [3]. Thus κ , A_0 , and C are elementary Γ -multiples of one another, and a transcendence statement for any one of them is a transcendence statement for all three.

Remark 1 (Notation). We write κ for this constant throughout. The symbol collides cosmetically with the Stokes constant “ K ” of an unrelated companion family (quadratic-growth V -family, Painlevé V on the surface $D_5^{(1)}$); the two are different numbers on different Sakai surfaces, and the present κ is fixed once and for all by (1)–(2).

1.2. Results and their evidential status. We organize the paper by a four-class evidence discipline, stated here so the reader can audit each assertion against the kind of evidence that supports it. PROVEN is reserved for results carrying a machine-checked Lean 4 proof with an audited axiom cone; STRUCTURAL denotes a complete mathematical argument verified by hand and computer algebra but not formalized; VERIFIED denotes a reproducible high-precision or exact symbolic computation; CONJECTURED denotes an open statement. A grade table appears in §10. *This paper introduces no new PROVEN item*; the only machine-checked substrate in the program is the EBR-III Lean cores, cited as such (§11), and EBR-IV’s own finitary identities are flagged as Lean-core candidates (§12). The principal results are:

- (R1) *Reduction and local type.* A Borel–2 reduction sends G to a rank-2 core H_2 with singular set $\{0, \infty\}$, both irregular of slope $\frac{1}{2}$; $\text{rig}(H_2) = 0$ (non-rigid, moduli dimension 2), and the ramification data select the Sakai surface $D_8^{(1)} = P_{\text{III}}(D_8)$ (§2–§3; VERIFIED).
- (R2) *Galois group.* $G_{\text{Gal}}(H_2) = \text{SL}_2$; in particular H_2 is irreducible with no Liouvillian solutions (§4; STRUCTURAL).
- (R3) *Constructive exponential period.* On the convergent Borel–2 transform, κ equals the connection coefficient A_{Φ} of an order-4 operator with an irregular point of slope $\frac{1}{4}$, computed to 129 digits by a monodromy spectral projector and confirmed by two further independent channels; κ lies in the ring of exponential periods of the connection (§5; STRUCTURAL, with VERIFIED numerics).
- (R4) *Monodromy point and coverage.* The point is $(\text{tr } M_0, \kappa)$ with $\text{tr } M_0 = -51.0655631399546\dots$ (irreducible); the solved D_8 connection problems are tau-side and do not compute the Lax-side κ (§6; VERIFIED).

- (R5) *Nulls.* Two positive-control-validated integer-relation batteries find κ neither elementary- Γ nor Barnes–Glaisher log-linear, to high precision (§7; VERIFIED).
- (R6) *Off-locus and the conditional theorem.* $(\text{tr } M_0, \kappa)$ lies off the algebraic $P_{\text{III}}(D_8)$ locus; under three named hypotheses and a verified datum, $\kappa \notin \overline{\mathbb{Q}}$ (§8–§9; VERIFIED off-locus, CONJECTURED-conditional theorem).

1.3. What we do not claim, and the ceiling. We are explicit about the limits of the method, since the temptation to overread a large Galois group or a non-rigid surface is precisely the error the EBR program guards against.

Non-rigidity (rig = 0, moduli dimension 2) does not imply that κ is transcendental; a large differential Galois group (SL_2) does not imply that κ is transcendental. The differential side targets the group only. The transcendence conclusion of §9 is licensed solely by the conjunction of its four named hypotheses, two of which are external, well-defined conjectures.

The discipline cuts in both directions, and we state the ceiling once here in the form it keeps throughout (it reappears boxed at §9):

The transcendence of C —equivalently of $\kappa = \Gamma(4/3) C / \sqrt{\pi}$ —remains conjectural unconditionally; it is not established at any grade in this paper. Exhibiting κ as a constructive exponential period, the off-locus verdict, the not-covered survey result, and the two integer-relation nulls prove nothing about transcendence; a closed form, had one fired, would have argued the opposite—elementarity in an extended class. The relevant period conjectures see the exponential provenance of κ , not the type of its singularity.

Accordingly the abstract and this introduction make no transcendence claim of any kind, no unqualified statement that κ or C “is a period”, and no claim that the group or the surface type forces transcendence.

2. REDUCTION TO A RANK-2 CORE AND THE BOREL-2 OPERATOR

2.1. From the continued fraction to the core H_2 . With $b(n) = 3n^2 + n + 1$ the numerators satisfy $Q_n = (3n^2 + n + 1)Q_{n-1} + Q_{n-2}$, $Q_0 = 1$, $Q_1 = 5$. The ordinary generating function $y(t) = \sum_{n \geq 0} Q_n t^n$ satisfies the inhomogeneous equation

$$3t^3 y'' + 10t^2 y' + (t^2 + 5t - 1)y = -1, \quad (3)$$

verified against the series to high order. Its homogeneous part is the rank-2 operator

$$H_2 = 3t^3 D^2 + 10t^2 D + (t^2 + 5t - 1), \quad D = \frac{d}{dt}, \quad (4)$$

the *core* of the EBR-IV analysis [CC3-2-CORE-CHAIN].

2.2. The Borel-2 operator and the home of κ . The series y is Gevrey of order 2 (the singular point $t = 0$ is irregular of slope $\frac{1}{2}$, §3), so its natural resummation passes through the *Borel-2 transform*

$$\Phi(z) = \sum_{n \geq 0} \frac{Q_n}{(n!)^2} z^n, \quad (5)$$

which is convergent (radius $1/3$) and holonomic. Writing $\theta = z d/dz$, the order-4 operator annihilating Φ is

$$L = \theta^2(\theta - 1)^2 - z(3\theta^2 + 7\theta + 5)\theta^2 - z^2, \quad (6)$$

equivalently $p_4\Phi^{(4)} + \dots + p_0\Phi = 0$ with $p_0 = -z^2$, $p_1 = -15z^2$, $p_2 = -z^2(47z - 2)$, $p_3 = -z^3(25z - 4)$, $p_4 = -z^4(3z - 1)$; L annihilates the series (5) to the orders tested (residual zero) [CC3-S6-PMAT]. Its finite singular set is $\{0, 1/3\}$, with indicial roots $\{0, 0, 1, 1\}$ at $z = 0$ and $\{-\frac{4}{3}, 0, 1, 2\}$ at $z = 1/3$, and *the point $z = \infty$ is irregular of slope $\frac{1}{4}$* . The last fact is decisive for the arithmetic home of the problem:

because L is globally irregular, its connection coefficients are exponential (rapid-decay) periods in the sense of the irregular Hodge / exponential-motive theory, not classical Kontsevich–Zagier periods.

The singularity at $z = 1/3$ corresponds, under $z = s/4$ and $s = R(\cdot)$, to the Borel-plane (instanton-action) singularity of the Gevrey-2 series; thus κ is intrinsically a *Stokes-type* constant, refining the EBR-II reading of A as a two-regular-point connection coefficient. By the transfer theorem of singularity analysis [13], κ is the $(1 - 3z)^{-4/3}$ -amplitude A_Φ of Φ at $z = 1/3$, whence the exact $\kappa = \Gamma(4/3)A_0$ of (2) [CC3-2S2-KAPPA-RES].

3. LOCAL STRUCTURE AND THE PAINLEVÉ III(D_8) SURFACE

3.1. Formal data and non-rigidity. In trace-normal form H_2 becomes $Y'' = rY$ with

$$r(t) = \frac{1}{3t^3} - \frac{5}{9t^2} - \frac{1}{3t}, \quad (7)$$

whose only singular points are $t = 0$ and $t = \infty$, *both irregular of slope $\frac{1}{2}$, ramification 2* [CC3-2-CORE-LOCAL]. The index of rigidity, computed for two slope- $\frac{1}{2}$ ramified points, is

$$\text{rig}(H_2) = (2 - 2) \cdot 4 + (1 + 1) - (1 + 1) = 0, \quad (8)$$

so H_2 is *non-rigid* with a 2-dimensional moduli space (a Painlevé phase space) and a single accessory parameter; the control families Airy, ${}_2F_1$, Bessel return $\text{rig} = 2$ (rigid) [CC3-2S2-RIG], [CC3-2-CONV-1].

3.2. The surface-type selector and the $D_8^{(1)}$ label. We assign the Sakai surface type by an explicit selector, computed from the local formal data, and we state it here because the label is load-bearing.

***Surface-type selector (RULE S).** For a non-rigid rank-2 connection with two irregular singular points, the Sakai surface type is determined by the multiset of local ramification orders. With both legs ramified, the dictionary $(1,1) \mapsto D_6^{(1)}$, $(2,1) \mapsto D_7^{(1)}$, $(2,2) \mapsto D_8^{(1)}$ is a bijection onto the degenerate Painlevé III surfaces. The type is computed from the formal data; it is not assigned by matching a solution against a catalogue.*

For H_2 the legs are $(2,2)$, so RULE S selects the Sakai surface $D_8^{(1)}$, i.e. $P_{\text{III}}(D_8)$, the surface with symmetry sublattice of type $D_8^{(1)}$ in Sakai's classification [4]; the assignment is consistent with $\text{rig} = 0$ and moduli dimension 2, and with the van der Put–Saito treatment of irregular rank-2 moduli [5]. An independent Padé screen of the second leg reproduces the singular value $z = 1/3$ to 34 digits with a converging pole table, corroborating the identification numerically [CC3-2S2-RULES]. This is the first VERIFIED Painlevé-surface label in the program; throughout, “the D_8 surface” means precisely this selector output.

4. THE DIFFERENTIAL GALOIS GROUP

4.1. $G_{\text{Gal}}(H_2) = \text{SL}_2$. Because H_2 has trivial determinant character, its Galois group lies in SL_2 . We exclude the three proper possibilities by Kovacic's algorithm [12], treating H_2 in its reduced form (7):

- *Case 1* (a rational exponential solution) is excluded: the only finite singular point $t = 0$ carries a half-integer indicial datum, and the associated Riccati equation has no solution in $\mathbb{C}(t)$.
- *Case 3* (finite imprimitive image) is excluded by the irregular slope $\frac{1}{2}$ at both points.
- *Case 2* (dihedral / imprimitive over a quadratic cover) is the only delicate one. The unique quadratic ramified cover that could split the local data is $t = x^2$, under which (7) pulls back to $R(x) = \frac{4}{3x^4} - \frac{53}{36x^2} - \frac{4}{3}$, with a single finite pole at $x = 0$ of order 4. The Riccati equation is then forced into the form $a/x^2 + b/x + c$ with $a^2 = \frac{4}{3}$; an exhaustive search over the field $\mathbb{Q}(\sqrt{3})$ generated by the local data is *empty*.

Hence no case applies and

$$G_{\text{Gal}}(H_2) = \text{SL}_2, \tag{9}$$

in particular H_2 is irreducible and has no Liouvillian solutions [11] [CC3-2-KOV]. The reduced form exhibits a symmetric degenerate confluent Heun (two rank-1 irregular points after $t = x^2$) structure [CC3-2-NF].

4.2. The discipline line, first statement. Equation (9) is a statement about the *group*. A Zariski-dense SL_2 does not, by itself, bound the transcendence degree of any single connection entry: connection coefficients are not differential-Galois invariants. The role of (9) in the conditional theorem (§9) is to supply, through a separate comparison, a *lower bound on the dimension of a motivic group*; it does nothing on its own. We return to this at hypothesis (H2).

5. κ AS A CONSTRUCTIVE EXPONENTIAL PERIOD

5.1. The connection coefficient A_Φ . The rank-2 solution y is divergent at $t = 0$, so the rapid-decay period pairing is realised on the convergent Borel–2 transform Φ of (5), governed by L of (6). At $z = 1/3$ the local exponents are $\{-\frac{4}{3}, 0, 1, 2\}$; the three integer exponents form a unipotent block killed by $(M - I)^3$, where M is the local monodromy, while the $-\frac{4}{3}$ direction is the eigenvalue $\mu = e^{-2\pi i/3}$. The *monodromy spectral projector*

$$A_\Phi \vec{s} = \frac{(M - I)^3 \vec{\varphi}}{(\mu - 1)^3}, \quad (M - I)^3 \vec{\varphi} = \vec{\varphi}_3 - 3\vec{\varphi}_2 + 3\vec{\varphi}_1 - \vec{\varphi}_0, \quad (10)$$

extracts the exponent- $(-\frac{4}{3})$ amplitude of the origin-analytic solution Φ by continuing *only* Φ through three loops around $z = 1/3$. The result,

$$A_\Phi = 1.539494848576641034843781903384 \dots = \kappa, \quad (11)$$

agrees with (1) to 129 digits; the deformation-invariance and four-component consistency witnesses reach 204 and 208 digits respectively, and A_Φ is real to $\sim 2 \times 10^{-209}$ [CC3-S6-PMAT]. (The working precision is set before any module-level constant is created; see the methodology note of §6.2.)

5.2. The κ -bridge. That A_Φ is exactly κ , and not merely numerically close, follows from an explicit Stokes-from-periods bridge, every factor stated. The gauge chain of §6 sends H_2 to the companion system $B = \begin{bmatrix} 0 & 1 \\ R & 0 \end{bmatrix}$ by a *scalar* gauge $y = x^{-17/6}w$; a scalar gauge leaves the Stokes *matrices* invariant, so the single off-diagonal multiplier $s_*(B; x = 0)$ of the slope- $\frac{1}{2}$ point equals $s_*(H_2; t = 0)$, which for a slope- $\frac{1}{2}$ point is the Borel-plane connection coefficient A_Φ . Combining with the transfer identity $A_\Phi = \Gamma(\frac{4}{3})A_0$,

$$\kappa := \Gamma(\frac{4}{3})A_0 = A_\Phi = s_*(B; x = 0), \quad (12)$$

the factor between s_* and κ being the identity and that between A_Φ and the large-order amplitude A_0 being exactly $\Gamma(\frac{4}{3})$; the identities hold to 130, 200, and 130 digits respectively [CC3-S6-CLOSE]. We record the consequence at the grade it earns:

Proposition 2 (STRUCTURAL). *κ lies in the ring of exponential periods of the H_2/L connection, constructively: it is the Borel-side connection coefficient A_Φ of (10), certified by the explicit integrand chain $y \rightsquigarrow \Phi = y \odot I_0(2\sqrt{z}) \rightsquigarrow A_\Phi$ together with the numerical coefficient (11).*

The integrand chain uses the modified Bessel kernel $I_0(2\sqrt{z})$, an exponential period by [15, 10.9.19], paired against the rapid-decay cycle in the sense of Hien’s duality [14]. The full exponential-*motive* membership (an abstract object of the Fresan–Jossen category with a realisation comparison) is not asserted here; it is hypothesis (H2)/(H4) of §9 and gap G2/G4 of §9.6.

5.3. Three independent channels. The value (11) is reached by three channels with disjoint inputs, which we separate because their mutual independence is the evidential backbone of (R3):

- *Channel A* (large-order): $A_0 = \lim_n Q_n / ((n!)^2 3^n n^{1/3})$ from the integer sequence Q_n alone ($N_{\max} = 60000$), ~ 60 digits, with purely $1/n$ corrections (no $n^{-1/3}$, no logarithm), confirming a simple dominant Borel singularity.
- *Channel B* (frozen composition): $\kappa = \Gamma(\frac{4}{3})C_{\text{EBR}}/\sqrt{\pi}$ from the EBR-III 169-digit C_{EBR} .
- *Channel C* (monodromy projector): (10), 129 digits, using only analytic continuation of Φ .

Channel A uses asymptotics of the original sequence, Channel B an independent high-precision computation of C_{EBR} , and Channel C the local monodromy of L ; they share no intermediate quantity, and they agree [CC3-2S2-KAPPA-NUM].

6. THE MONODROMY POINT AND THE COVERAGE VERDICT

6.1. The point ($\text{tr } M_0, \kappa$). The topological monodromy of the trace-normal connection (7) about $t = 0$ has trace

$$\text{tr } M_0 = -51.06556313995466226983167460994566156792\dots, \quad (13)$$

a hyperbolic SL_2 element ($|\text{tr } M_0| \gg 2$), so the monodromy is *irreducible* and the point lies off every reducibility locus. With κ , this fixes the 2-dimensional D_8 character-variety point $(\text{tr } M_0, \kappa)$ [CC3-2S2-2A-COORDS]. The accuracy witness is the invariance of (13) under deformation of the integration path, radius, and node count (radius-independent to ~ 157 digits, phase to ~ 177), *not* the structural relation $\det = 1$.

6.2. A methodology note. Two integrator defects were found and fixed in computing (13), and we record them because each produces a plausible but wrong stable value. First, an asymmetric loop range silently dropped a node. Second—a named hazard for the whole corpus—module-level `mpf` constants created before the working precision is raised are evaluated at the default 15 digits, which can counterfeit a 141-digit “stability”. The rule adopted throughout is to set the precision before creating any constant [CC3-2S2-2A-METH].

6.3. The gauge dictionary and the coverage verdict. The scalar gauge chain of §5.2 carries H_2 to the companion B with two rank-1 irregular points—the D_8 shape, a symmetric degenerate confluent Heun system [CC3-2S2-2B-DICT]. We then surveyed the solved D_8 connection problems to ask whether any computes κ . The verdict is *not covered*, and the reason is a distinction we keep explicit in every crossing sentence:

The tau-side / Lax-side line. The constant κ is a Lax-side Stokes multiplier of the linear system. The solved D_8 results compute tau-side objects—functions on the monodromy manifold—taking the monodromy data as input.

Concretely: Its–Lisovyy–Prokhorov compute the connection *constant* of a Painlevé tau function as an explicit Barnes- G/Γ product of the monodromy data [7]; Gavrylenko–Lisovyy represent the isomonodromic tau function as a Fredholm determinant assembled from the monodromy data [8]. In both, the monodromy (the analogue of κ) is the *input*, never the output; neither yields a closed form for a Stokes multiplier. Moreover H_2 is non-rigid (8), hence outside the rigid hypergeometric/Bessel/Airy catalogue where Stokes data are Γ -quotients [9, 10]. We therefore record (III) NOT COVERED: no stated theorem computes the coordinate κ at our point [CC3-2S2-2C-VERDICT]. By the ceiling, this is not evidence for transcendence; it is the absence of a closed form that would have argued against it.

7. TWO INTEGER-RELATION NULLS

We record two high-precision exclusions, each positive-control-validated, and the discipline rule that governs their reading.

7.1. The elementary null. To 169 digits, C_{EBR} (hence κ , by (2)) is not an elementary Γ -quotient on the reflection-normalized $\frac{1}{24}\mathbb{Z}$ grid, not a low-height combination of $\{1, \pi, \log 2, \log 3, \gamma_E, \text{Catalan}, \zeta(3), \zeta(5)\}$, and not algebraic of degree ≤ 8 at height $\leq 10^6$; the positive controls fire [3] [EBR3-B-GAMMA] [EBR3-B-CONST] [EBR3-B-ALG]. This is the EBR-III frozen battery, cited and not recomputed.

7.2. The Barnes–Glaisher log-space null. In log space, neither $\log \kappa$ nor $\log A_0$ is a $\mathbb{Z}$ -linear combination over a Barnes- G /Glaisher-extended basis with arguments j/m , $m \in \{2, 3, 4, 6, 12\}$, to height $\leq 10^{12}$ at 150 digits; the positive control $G(1/2) = 2^{1/24}\pi^{-1/4}e^{1/8}A^{-3/2}$ fires with coefficient vector $[24, -1, 6, 36, -3]$ [CC3-2S2-3-BARNES].

7.3. The null-discipline rule. Both nulls are read under a fixed detection-threshold rule, which we state because it is twice load-bearing in the program and once caught a live false positive.

Null-discipline. An integer-relation search among n terms with coefficient height $\leq H$, tolerance $10^{-\tau}$, and working precision d detects a genuine relation only if $\tau \gg n \log_{10} H$ and $d > \tau$; otherwise any relation returned is a precision artifact, recognizable because its

coefficients have magnitude $\sim 10^{\tau/n}$. A null under a well-resourced search is a real exclusion bound to the stated height.

In the locus computation of §8 a first-pass degree-10 search at $\tau = 60$ returned coefficients of size $\sim 6 \times 10^5 = 10^{60/10}$, the artifact signature exactly; re-resourcing to $\tau = 72$ with $(\deg + 1) \log_{10} H \leq 56$ and $d = 85$ returned the correct null [EBR4-METH-NULLDISC]. By the ceiling, a null bounds non-elementarity to the searched precision and says nothing about $\overline{\mathbb{Q}}$.

8. OFF THE CLASSICAL LOCUS

The classical/algebraic $P_{\text{III}}(D_8)$ locus has two strata: **(R)** reducible monodromy (Riccati / special-function solutions) and **(A)** algebraic solutions (finite braid orbit, algebraic trace coordinates). We exclude both at our point.

Stratum (R), unconditional. By (9) the Galois group is Zariski-dense in SL_2 , so the connection is irreducible and lies in no Borel subgroup; there is no reducible/Riccati member to be, and no numerical margin is needed [EBR4-1-LOCUS-DIRECT].

Stratum (A), well-resourced. For $P_{\text{III}}(D_8)$ the only algebraic solutions are $q = c\sqrt{t}$ with $c^4 = 1$, a set of at most four solutions defined over $\mathbb{Q}(i)$ [6]. The absolute Galois group permutes the four c 's, so the corresponding character-variety images form a Galois-stable set of size ≤ 4 , and each coordinate—in particular the trace—is an algebraic number of degree ≤ 4 over $\mathbb{Q}$ whose conjugates are themselves trace coordinates of the finite orbit (hence of bounded modulus and small height). Now $\text{tr } M_0$ of (13) is *not* algebraic of degree ≤ 4 at height $\leq 10^{10}$: the PSLQ search is null and well-resourced ($\tau = 72 \gg 5 \log_{10} 10^{10} = 50$, reinforced at degree ≤ 6 , $H \leq 10^8$ and degree ≤ 8 , $H \leq 10^6$), with the controls $\sqrt{2}$, $2 \cos(2\pi/7)$, $2 \cos(2\pi/5)$ all firing. The degree- ≤ 4 test strictly dominates the orbit degree bound, and independently M_0 has infinite order ($|\lambda| = 0.0196 \neq 1$), excluding the finite-order subcase outright [EBR4-1-LOCUS-DIRECT]. Therefore $(\text{tr } M_0, \kappa)$ is not a $q = c\sqrt{t}$ point, and the point lies off the locus.

Remark 3 (the residual, stated honestly). Stratum (A) is *hardened, not closed*. The argument assumes the orbit trace coordinate has height $\leq 10^{10}$ —overwhelmingly safe, since it is a bounded-modulus algebraic integer of degree ≤ 4 , but not proven here. The remaining step is a finite, elementary computation: the explicit $q = c\sqrt{t}$ Stokes/monodromy trace. We state it as an open sub-problem (§12, item 3), not as a completed exclusion.

9. THE CONDITIONAL THEOREM

9.1. The named conjecture. The transcendence input is the *period conjecture for exponential motives* of Fresan–Jossen, the exponential analogue of Grothendieck's: for an exponential motive M ,

$$\text{trdeg}_{\mathbb{Q}} \langle \text{periods}(M) \rangle = \dim G_{\text{mot}}(M), \quad (14)$$

every polynomial relation among the exponential periods being of motivic origin [16, 17, 19]. (An AI-suggested arXiv identifier for this statement was checked and found to point to an unrelated physics paper; we cite the work as a book in preparation and attach no preprint number.)

9.2. The differential-to-motivic comparison. The bridge from (9) to a motivic dimension is a separate statement from (14), and we make it explicit because conflating the two would overstate the result. For a motive whose de Rham realisation carries the relevant connection, the differential Galois group injects into the motivic Galois group as (the identity component of) a normal subgroup, whence

$$\dim G_{\text{mot}}(M) \geq \dim G_{\text{Gal}} = \dim \text{SL}_2 = 3. \quad (15)$$

This is theorem-grade in the classical (pure, regular-singular) setting [18, 19, 20]. Here M is an *exponential* motive and the connection L is irregular, so (15) is applied within the Fresan–Jossen framework and is contingent on realising M as a genuine object of that category with a Stokes-aware comparison—hypothesis (H2)/(G4). We stress that (15) and (14) are *logically independent*: (14) converts a dimension into a transcendence degree but never bounds the dimension from below, while (15) bounds the dimension but carries no transcendence content. Dropping either leaves the conclusion unreachable, as the drop-one analysis below records [EBR4-0-HYP].

9.3. The theorem. We state the result verbatim in its audited four-hypothesis form.

Theorem 4 (conditional). *Let M be the exponential motive attached to the rank-2 connection H_2 (realised on the convergent Borel-2 transform Φ ; order-4 operator L , singular $\{0, 1/3\}$, $z = \infty$ irregular of slope $\frac{1}{4}$). Assume:*

(H1, STRUCTURAL) κ is an exponential period of M —constructively the connection coefficient $A_\Phi = \kappa$ (Proposition 2, 129 digits).

(H2, MOTIVIC / VERIFIED differential) $\dim G_{\text{mot}}(M) \geq 3$, via $G_{\text{Gal}}(H_2) = \text{SL}_2$ (exact Kovacic, dimension 3) transported to the motivic group by the comparison $\dim G_{\text{mot}} \geq \dim G_{\text{Gal}}$ (15) (theorem-grade classically [18, 19]; here contingent on the (G4) realisation and the irregular/exponential form of the comparison).

(H3, CONJECTURED external) the Fresan–Jossen period conjecture (14) for exponential motives.

(H4, VERIFIED) the period-count / non-degeneracy data: $\dim H_{\text{dR}}^1(H_2) = 2$, the connection is irreducible (SL_2 Zariski-dense) and lies off the algebraic $P_{\text{III}}(D_8)$ locus, and κ is the non-vanishing off-diagonal connection entry ($A_\Phi \neq 0$), not a diagonal normalisation factor.

Then $\text{trdeg}_{\mathbb{Q}}(\text{periods}(M)) = \dim G_{\text{mot}}(M) \geq 3$ and the off-diagonal entry κ is not algebraic: $\kappa \notin \overline{\mathbb{Q}}$. Consequently the EBR-I/EBR-II connection coefficient $C = \kappa\sqrt{\pi}/\Gamma(4/3)$ is transcendental as well (an elementary Γ -multiple of κ).

The conclusion is licensed only under the full conjunction $\text{H1} \wedge \text{H2} \wedge \text{H3} \wedge \text{H4}$ (drop-one analysis: each hypothesis is necessary). Dropping H3 leaves the

unconditional residue only: κ is a non-classical exponential period off the algebraic $P_{\text{III}}(D_8)$ locus—which says nothing about $\overline{\mathbb{Q}}$.

Discussion of necessity (drop-one). Drop H1: if κ is not known to be a period of M , then (14) constrains it not at all. Drop H2: if $\dim G_{\text{mot}}(M)$ may be small, (14) makes the period algebra possibly algebraic, and κ possibly algebraic; crucially (14) never supplies $\dim \geq 3$. Drop H3: then $\dim G_{\text{mot}}(M) \geq 3$ is a purely group-theoretic fact with no transcendence content. Drop H4: the transcendence could live entirely in other period entries while the specific entry κ is a diagonal normalisation. Each hypothesis is thus necessary, and H2, H3 are mutually non-implying [EBR4-0-HYP] [CC3-3-CONDITIONAL]. Positively, (H4) is what localises the algebra-level bound $\text{trdeg} \geq 3$ to κ itself: in the period-torsor formalism the motivic group acts on the de Rham / rapid-decay comparison, an SL_2 of dense image fixes no non-constant off-diagonal coordinate of the 2-dimensional realisation, and (H4) certifies κ to be such an entry (non-vanishing, off-diagonal, not a diagonal normalisation)—so κ is not in the fixed field $\overline{\mathbb{Q}}$. Without that certificate the dimension bound would not descend to the single entry. $\square$

9.4. Two facts about κ , never elided. Hypothesis (H4) and the conclusion concern the *same* entry κ , and a hostile reading could hear circularity. We therefore separate two facts that are genuinely distinct:

- (a) $\kappa = A_{\Phi}$ is the *identified, non-vanishing* off-diagonal connection entry: $A_{\Phi} \neq 0$ to 129 digits, structurally $\Gamma(\frac{4}{3})C_{\text{EBR}}/\sqrt{\pi}$ with every factor nonzero. This is VERIFIED and is an input (H4).
- (b) The *non-algebraicity* of that entry is what the conjectural chain (H1)–(H4) *delivers*; it is the conclusion, not an assumed datum.

The existence and non-vanishing of the entry are independent of the transcendence question: (a) is settled unconditionally by the period-matrix computation, while (b) requires the full conjunction. Membership in the ring of exponential periods is, by itself, no arithmetic constraint— $\overline{\mathbb{Q}}$ lies inside that ring—so (H1) does not pre-judge (b); the non-algebraicity is supplied only by the dimension lower bound (H2) feeding the period conjecture (H3). Conflating (a) and (b) would be the circularity; keeping them apart is the content of the theorem.

9.5. The ceiling, boxed.

Ceiling (final form, both directions). The transcendence of C , equivalently of $\kappa = \Gamma(4/3)C/\sqrt{\pi}$, remains conjectural unconditionally; Theorem 4 is the program’s ceiling and says so. Exhibiting κ as a constructive exponential period (Proposition 2), the off-locus verdict (§8), the not-covered survey (§6.3), and the two nulls (§7) prove nothing about transcendence; a closed form, had one fired, would have argued the opposite—elementarity in an extended class. The period conjectures see the exponential provenance of κ , not its singularity type.

9.6. The gap list. The conditional chain rests on three open inputs, which we make a first-class artifact (this is the open-problems core, sharpened by the hypothesis audit so that the conjectural mass sits in two named, external places):

Gap	Statement	What would close it
G1	The Fresan–Jossen period conjecture (14) is open.	A proof, or the single instance for M .
G2/G4	M is not yet built as a Fresan–Jossen-category exponential motive with the expected Galois group and a verified irregular realisation comparison (15); this is where the conjectural mass concentrates.	Construct $M = (X, f)$ and verify the Stokes-aware comparison; the seed is the $I_0(2\sqrt{z})$ kernel.
G5	The off-locus exclusion of stratum (A) uses a height- $\leq 10^{10}$ bound on the orbit trace coordinate (Remark 3).	The explicit $q = c\sqrt{t}$ Stokes/monodromy trace—a finite elementary computation.

The merge of the earlier “genericity” gap into G2/G4 is the substance of §9.4: entry-wise non-degeneracy is verified, so the only residual is the motivic realisation, not a separate non-vanishing hope [EBR4-1-LOCUS-DIRECT].

10. FOUR-CLASS GRADE TABLE

Result	Grade	Evidence
Core H_2 , Borel-2 operator L , slope- $\frac{1}{4}$	V/S	[CC3-2-CORE-CHAIN], [CC3-S6-PMAT]
∞		
$\{0, \infty\}$ both slope $\frac{1}{2}$; $\text{rig}(H_2) = 0$	V	[CC3-2-CORE-LOCAL], [CC3-2S2-RIG]
RULE $S \Rightarrow D_8^{(1)} = P_{\text{III}}(D_8)$	V	[CC3-2S2-RULES]
$G_{\text{Gal}}(H_2) = \text{SL}_2$; non-Liouvillian	S	[CC3-2-KOV]
$\kappa = \Gamma(\frac{4}{3})A_0$ exact; transfer	S	[CC3-2S2-KAPPA-RES]
$A_\Phi = \kappa$ to 129 d (projector); three channels	S/V	[CC3-S6-PMAT], [CC3-2S2-KAPPA-NUM]
κ a constructive exponential period (κ -bridge)	S	[CC3-S6-CLOSE]
$\text{tr } M_0 = -51.0655 \dots$ hyperbolic; dim-2 point	V	[CC3-2S2-2A-COORDS]
gauge dictionary $H_2 \rightarrow B$ (D_8 shape)	S	[CC3-2S2-2B-DICT]
coverage (III) NOT COVERED (tau vs Lax)	V	[CC3-2S2-2C-VERDICT]
elementary 169-d null	V	[EBR3-B-GAMMA], [EBR3-B-CONST], [EBR3-B-ALG]
Barnes–Glaisher log-space null	V	[CC3-2S2-3-BARNES]
null-discipline rule	V	[EBR4-METH-NULLDISC]
off the algebraic $P_{\text{III}}(D_8)$ locus (R uncond., A hardened)	V/S	[EBR4-1-LOCUS-DIRECT]
$H2 \perp H3$ audit; comparison explicit	V	[EBR4-0-HYP]
Conditional theorem: $\kappa \notin \mathbb{Q}$ under H1–H4	C	[CC3-3-CONDITIONAL]
Unconditional transcendence of C/κ	C	[CC1-DISCIPLINE]

No row is graded PROVEN: the program’s machine-checked cores are those of EBR-III (§11); EBR-IV’s finitary identities are Lean-core candidates (§12).

11. ON THE LEAN SUBSTRATE

EBR-IV adds no machine-checked core. The only PROVEN items in the program are the four EBR-III Lean 4 / Mathlib cores (degree-bound arithmetic, pullback-exponent integrality, eigenvalue-parity bookkeeping, and the parity of the relevant 4-cycle), with axiom cones audited to subsets of $\{\text{propxt}, \text{Classical.choice}, \text{Quot.sound}\}$ and no `sorryAx` [3, 21]. We flag the EBR-IV finitary identities that the same template could formalize in §12.

12. OPEN PROBLEMS

- (1) *Unconditional transcendence of C/κ (op:cc-transcendence)*. The ceiling; Theorem 4 stands on G1 and G2/G4, two named external conjectures.

- (2) *The motivic realisation* (G2/G4). Construct $M = (X, f)$ as a Fresan–Jossen object and verify the irregular realisation comparison (15); the seed is the $I_0(2\sqrt{z})$ kernel of §5.2.
- (3) *The full off-locus close* (G5). The explicit $q = c\sqrt{t}$ Stokes/monodromy trace removes the height- $\leq 10^{10}$ residual of Remark 3; this is a finite elementary computation.
- (4) *Lean-core candidates for EBR-IV* (op:cc3-5): the L -operator and indicial data, the $H_2 \rightarrow B$ gauge-chain identity, the Kovacic $\mathbb{Q}(\sqrt{3})$ -emptiness certificate, and the de Rham dimension counts, on the template of the EBR-III cores.
- (5) *A surface-type synthesis* (flagged). Both program constants—the Stokes constant of the quadratic-growth V -family on $D_5^{(1)}$ and the present κ on $D_8^{(1)}$ —are Stokes data of rank-2 Painlevé Lax operators. Whether the Sakai surface type on which a continued fraction’s growth constant lands predicts the arithmetic of that constant is a question larger than this paper; we record it as a pointer, not a result.

REPRODUCIBILITY

All computations are reproducible from the accompanying package (`repro_ebr4/`): the reduction and Borel-2 operator, the rigidity and RULE S screen, the Kovacic classification, the period-matrix/connection coefficient by the monodromy projector, the monodromy point, the coverage survey, the Barnes battery, the hypothesis-independence audit, and the hardened locus exclusion, each with a canonical content hash recorded in `HASH_MANIFEST.json`; a `README` gives a one-command verification path and `verify.py` re-derives every embedded self-hash and file hash (exit code 0). The three κ channels are independently reproducible. The frozen EBR-III package is untouched and is cited for the inherited C_{EBR} and the elementary null.

Load-bearing artifacts are pinned by SHA-256 prefix (full digests in `HASH_MANIFEST.json`): the period matrix / connection coefficient `cc3_s6_1_period_matrix_results.json` begins `56adcb10...`; the κ -bridge `cc3_s6_2_bridge_results.json` begins `2cc2f6fb...`; the hypothesis audit `ebr4_0_hypothesis_audit_results.json` begins `d86256cc...`; the hardened locus `ebr4_1_locus_hardening_results.json` begins `dae56db4...`; and the EBR-III integer-relation null begins `9a3f942d...`.

REFERENCES

- [1] Papanokechi, *An Edge–Borel Radius Theorem for Positive Polynomial Continued Fractions*, Zenodo, 10.5281/zenodo.20564079.
- [2] Papanokechi, *The EBR Amplitude as a Connection Coefficient: Characterization and a Rigidity Dividing-Line Conjecture*, Zenodo, 10.5281/zenodo.20566465.

- [3] Papanokechi, *The EBR operator at $d = 2$ is not a G -operator and its differential Galois group is $SL(4)$* , Zenodo (EBR-III); concept DOI to be pinned at deposit.
- [4] H. Sakai, *Rational surfaces associated with affine root systems and geometry of the Painlevé equations*, Comm. Math. Phys. **220** (2001), 165–229.
- [5] M. van der Put and M.-H. Saito, *Moduli spaces for linear differential equations and the Painlevé equations*, Ann. Inst. Fourier (Grenoble) **59** (2009), 2611–2667.
- [6] Y. Ohyama, H. Kawamuko, H. Sakai, and K. Okamoto, *Studies on the Painlevé equations V. Third Painlevé equations of special type $P_{III}(D_7)$ and $P_{III}(D_8)$* , J. Math. Sci. Univ. Tokyo **13** (2006), 145–204.
- [7] A. Its, O. Lisovyy, and A. Prokhorov, *Monodromy dependence and connection constants for Painlevé tau functions*, Comm. Math. Phys. **363** (2018), 1–58.
- [8] P. Gavrylenko and O. Lisovyy, *Fredholm determinant and Nekrasov sum representations of isomonodromic tau functions*, Comm. Math. Phys. **363** (2018), 1–58.
- [9] A. S. Fokas, A. R. Its, A. A. Kapaev, and V. Yu. Novokshenov, *Painlevé Transcendents: The Riemann–Hilbert Approach*, Math. Surveys Monogr. **128**, Amer. Math. Soc., 2006.
- [10] N. M. Katz, *Rigid Local Systems*, Ann. of Math. Stud. **139**, Princeton Univ. Press, 1996.
- [11] M. van der Put and M. F. Singer, *Galois Theory of Linear Differential Equations*, Grundlehren der mathematischen Wissenschaften **328**, Springer, 2003.
- [12] J. J. Kovacic, *An algorithm for solving second order linear homogeneous differential equations*, J. Symbolic Comput. **2** (1986), 3–43.
- [13] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge Univ. Press, 2009 (Ch. VI, singularity analysis / transfer).
- [14] M. Hien, *Periods for flat algebraic connections*, Invent. Math. **178** (2009), 1–22 (rapid-decay homology and the period pairing).
- [15] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, dlmf.nist.gov, §10.9.
- [16] J. Fresán and P. Jossen, *Exponential Motives* (book in preparation).
- [17] M. Kontsevich and D. Zagier, *Periods*, in: *Mathematics Unlimited—2001 and Beyond*, Springer, 2001, 771–808.
- [18] Y. André, *Pour une théorie inconditionnelle des motifs*, Publ. Math. IHÉS **83** (1996), 5–49.
- [19] Y. André, *Une introduction aux motifs (motifs purs, motifs mixtes, périodes)*, Panoramas et Synthèses **17**, Soc. Math. France, 2004.
- [20] Y. André, *Différentielles non commutatives et théorie de Galois différentielle ou aux différences*, Ann. Sci. École Norm. Sup. **34** (2001), 685–739.
- [21] The Mathlib Community, *The Lean Mathematical Library*, CPP 2020, 367–381.